

Name : Putra Amalul Kamal

Student ID : 001202300221

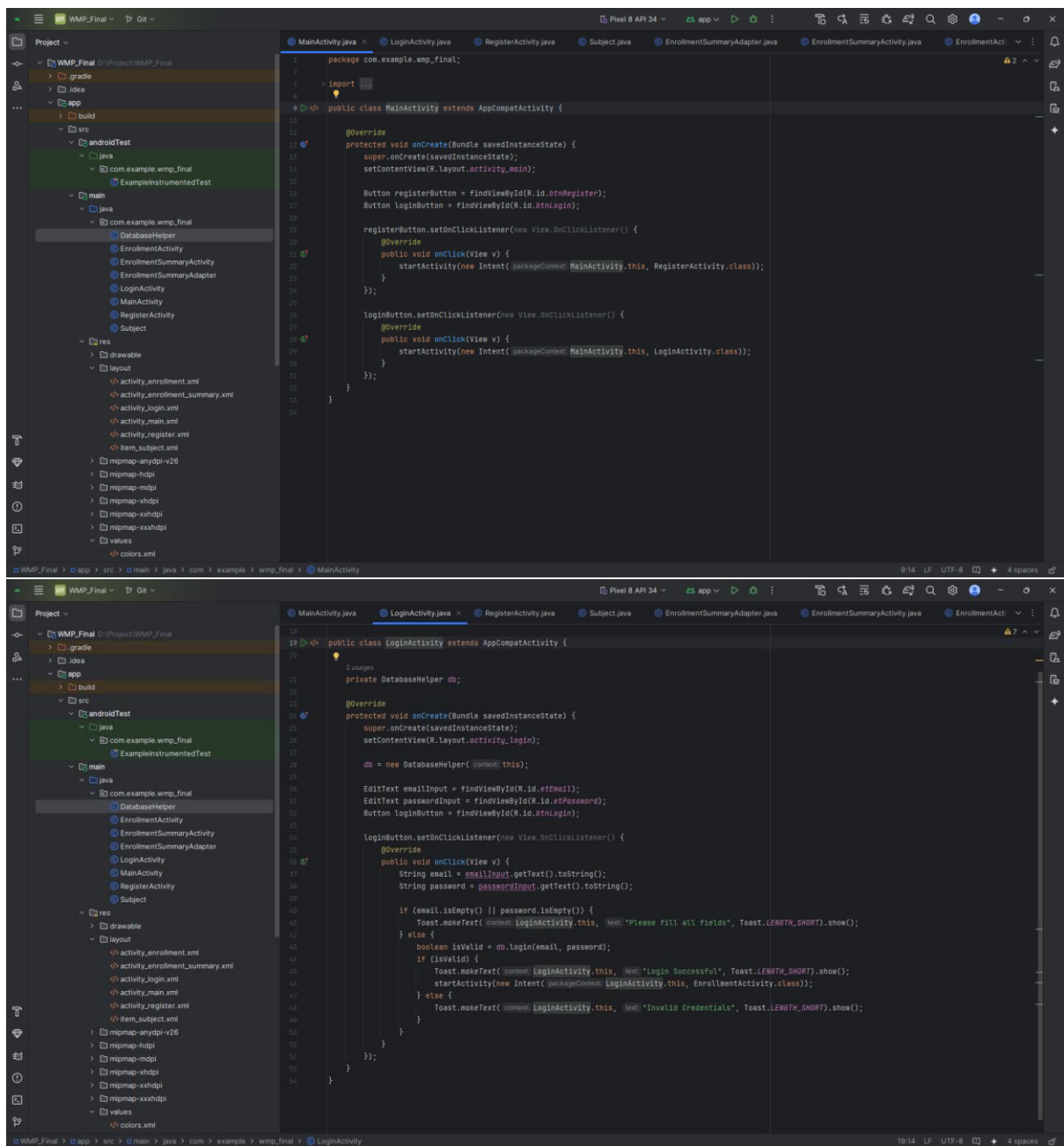
Class : IT 6

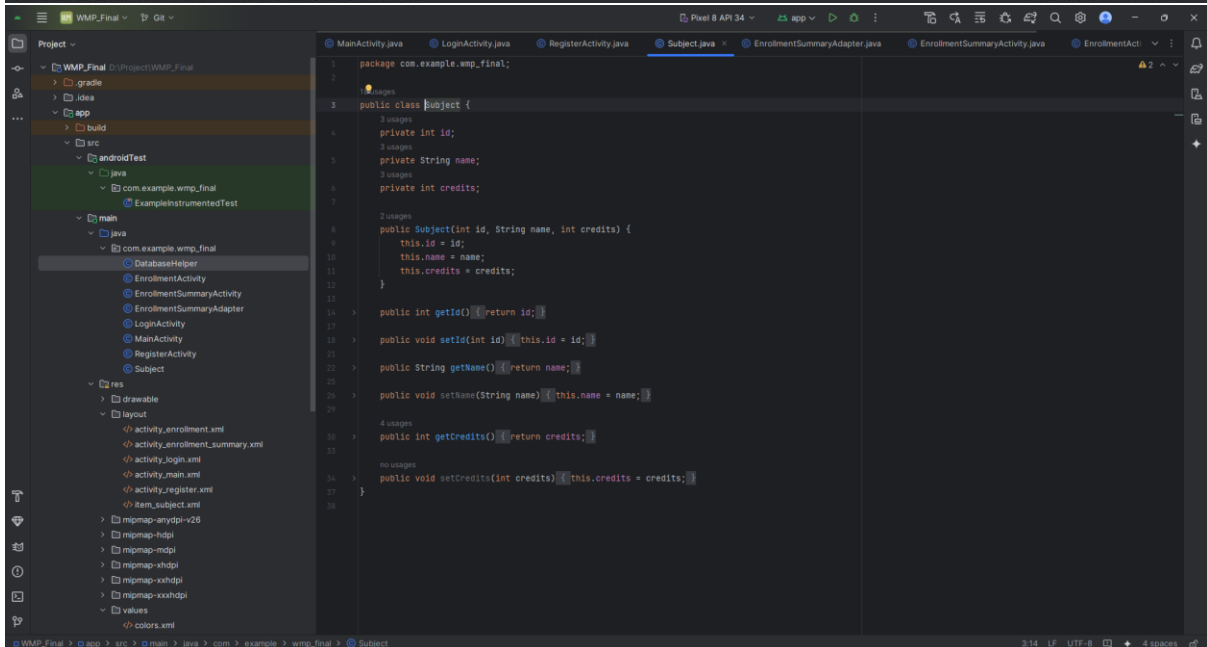
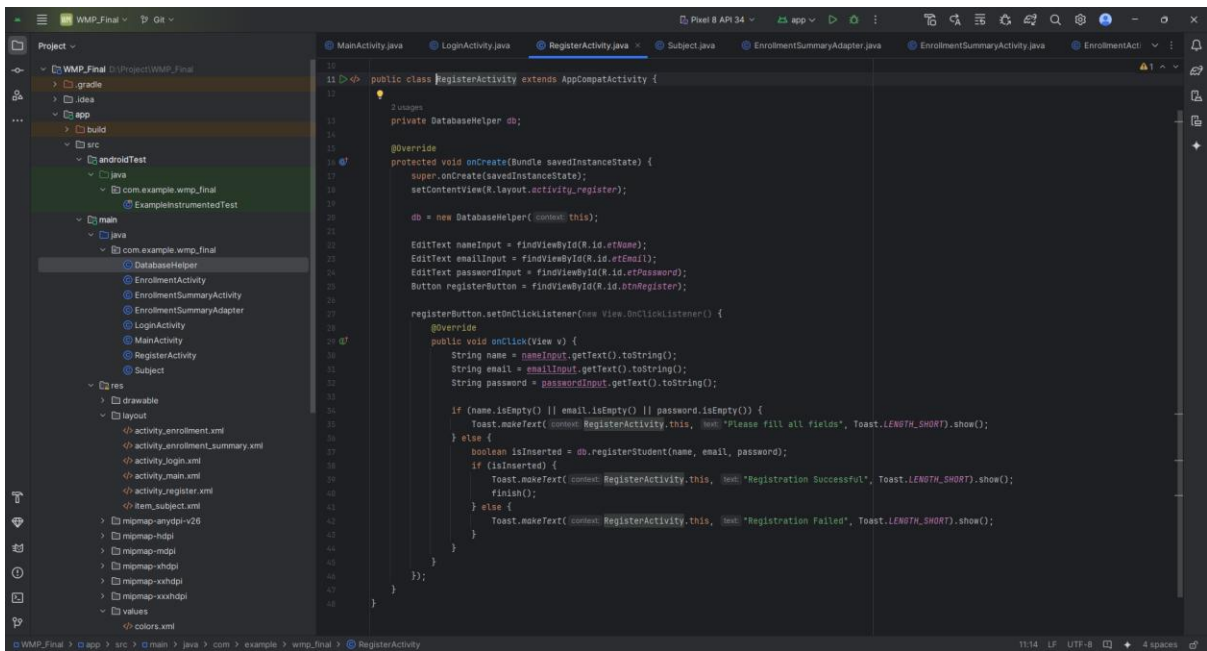
Final Exam – WMP

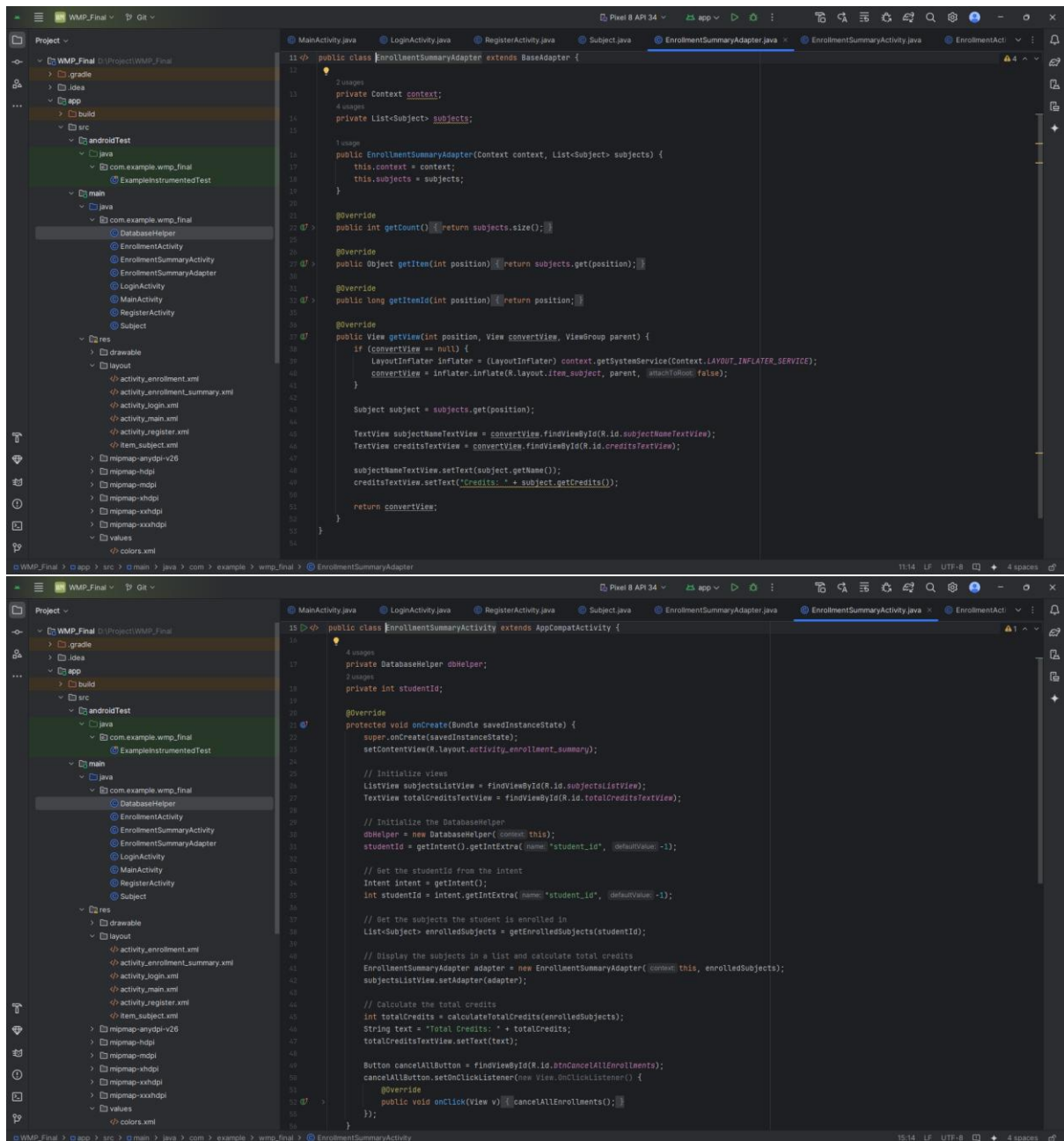
Create Android Application including the Database with the following requirements:

- User: student
- Features: register student, login, enrollment menu, select subject, enrollment summary (list of subjects and total credits)
- Additional: a max limit where total credits cannot more than x (i.e 24 credits)
- ERD

Screenshots





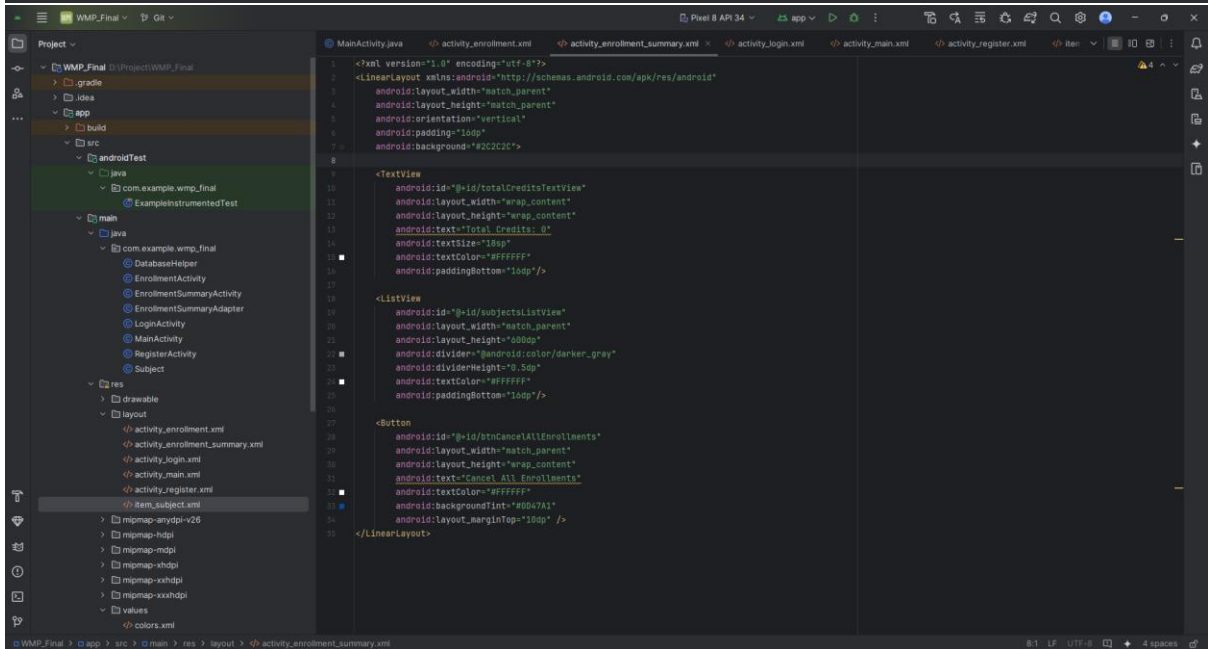
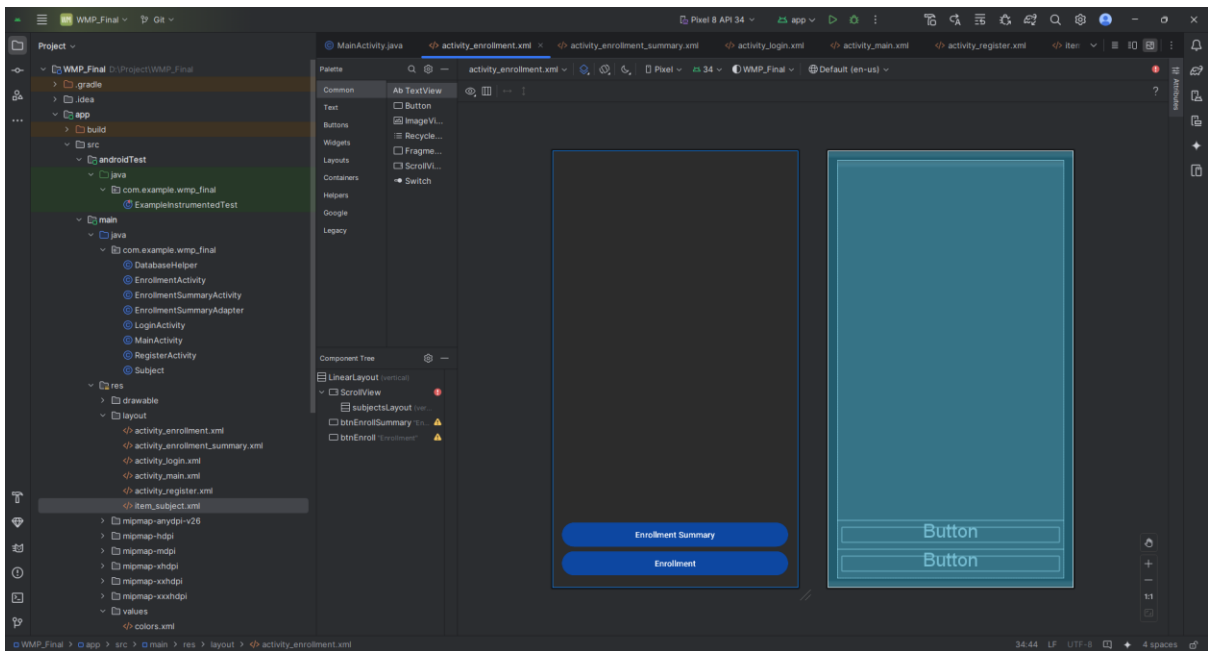


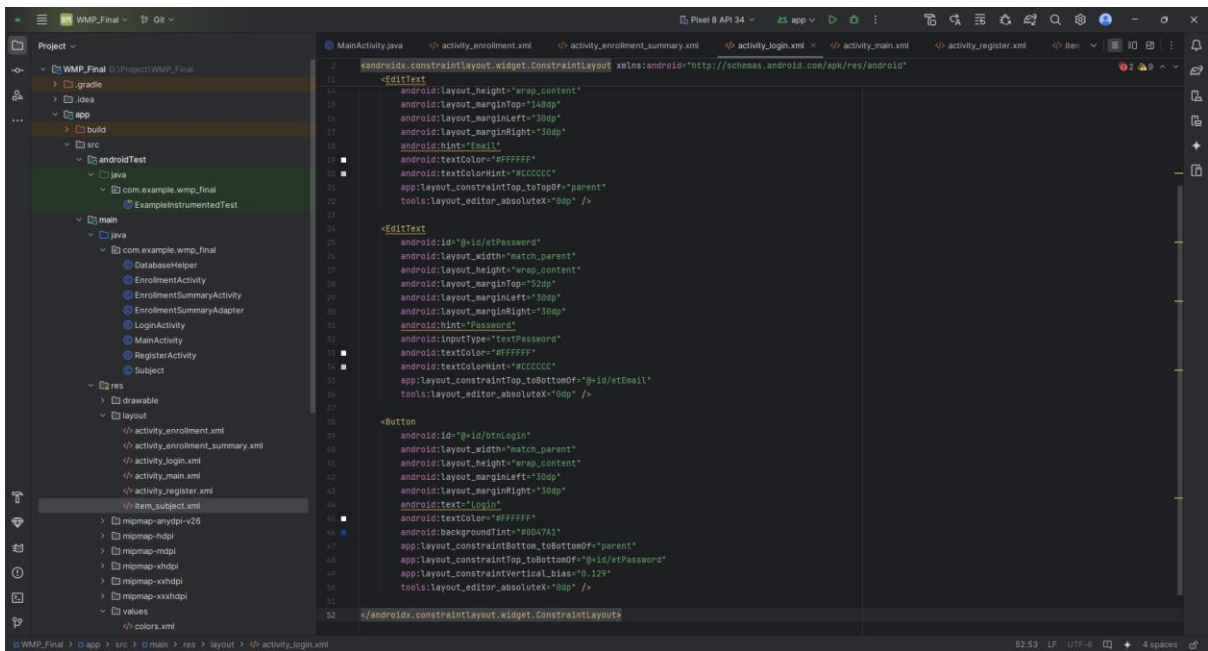
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Project
└─ WMP_Final
   └─ app
      └─ src
         └─ java
            └─ com.example.wmp_final
               └─ main
                  └─ java
                     └─ com.example.wmp_final
                        └─ EnrollmentActivity.java

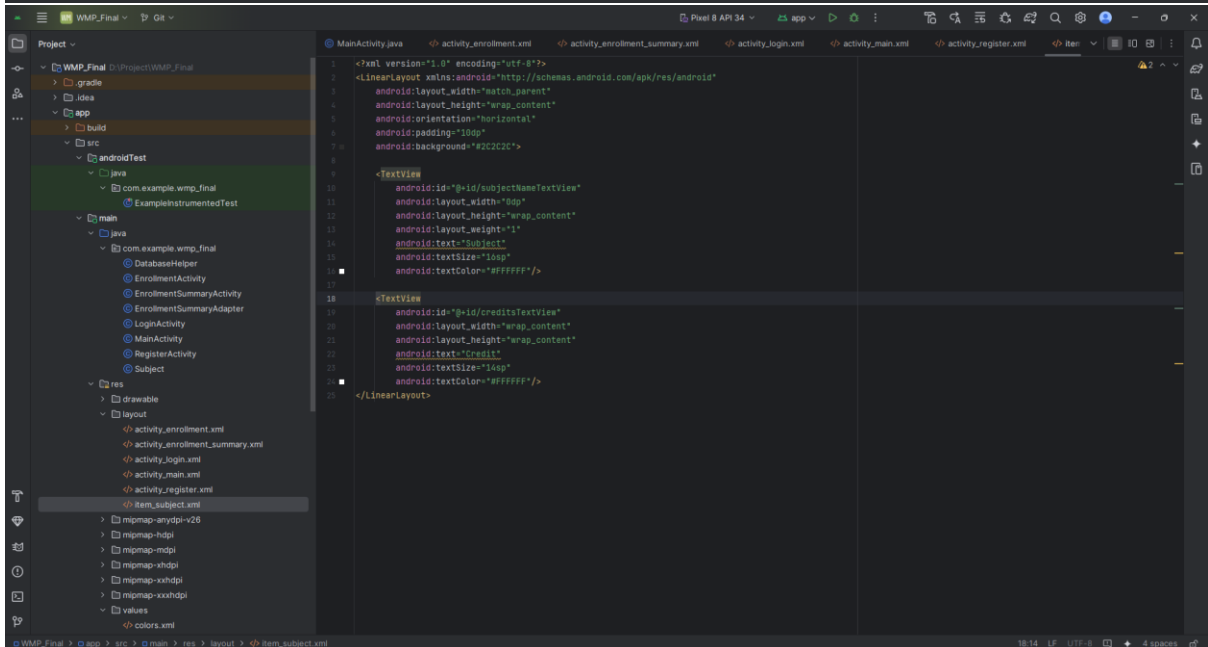
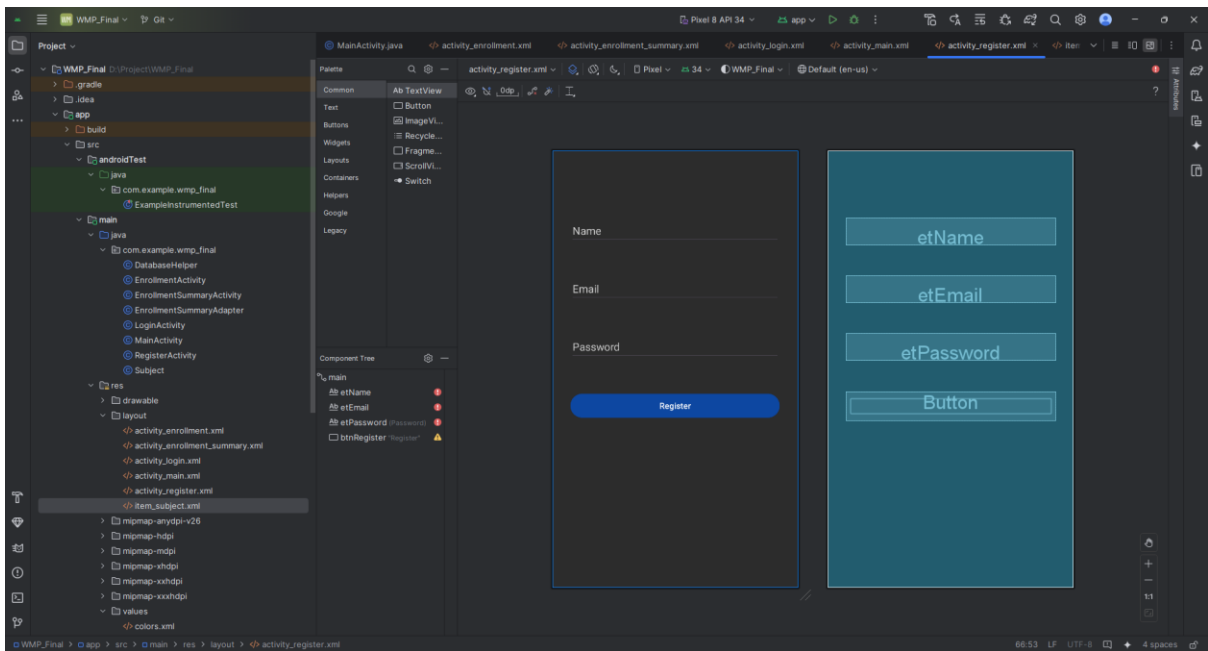
EnrollmentActivity.java
1 package com.example.wmp_final;
2
3 import androidx.appcompat.app.AppCompatActivity;
4
5 public class EnrollmentActivity extends AppCompatActivity {
6
7     5 usages
8     private DatabaseHelper db;
9     6 usages
10    private int studentId; // Pass this from LoginActivity
11    1 usage
12    private final int maxCredits = 24;
13
14    @Override
15    protected void onCreate(Bundle savedInstanceState) {
16        super.onCreate(savedInstanceState);
17        setContentView(R.layout.activity_enrollment);
18
19        db = new DatabaseHelper(this);
20        studentId = getIntent().getStringExtra("STUDENT_ID");
21
22        LinearLayout subjectsLayout = findViewById(R.id.subjectsLayout);
23        Button enrollButton = findViewById(R.id.btnEnroll);
24        Button enrollSummaryButton = findViewById(R.id.btnEnrollSummary);
25
26        List<Subject> subjects = db.getSubjects();
27        List<CheckBox> checkBoxes = new ArrayList<>();
28
29        for (Subject subject : subjects) {
30            CheckBox checkBox = new CheckBox(this);
31            String text = subject.getName() + " (" + subject.getCredits() + " credits)";
32            checkBox.setText(text);
33            checkBox.setTag(subject);
34            subjectsLayout.addView(checkBox);
35            checkBoxes.add(checkBox);
36        }
37
38        enrollButton.setOnClickListener(new View.OnClickListener() {
39            @Override
40            public void onClick(View v) {
41                List<Integer> selectedSubjects = new ArrayList<>();
42                int totalCredits = 0;
43            }
44        });
45    }
46}
```

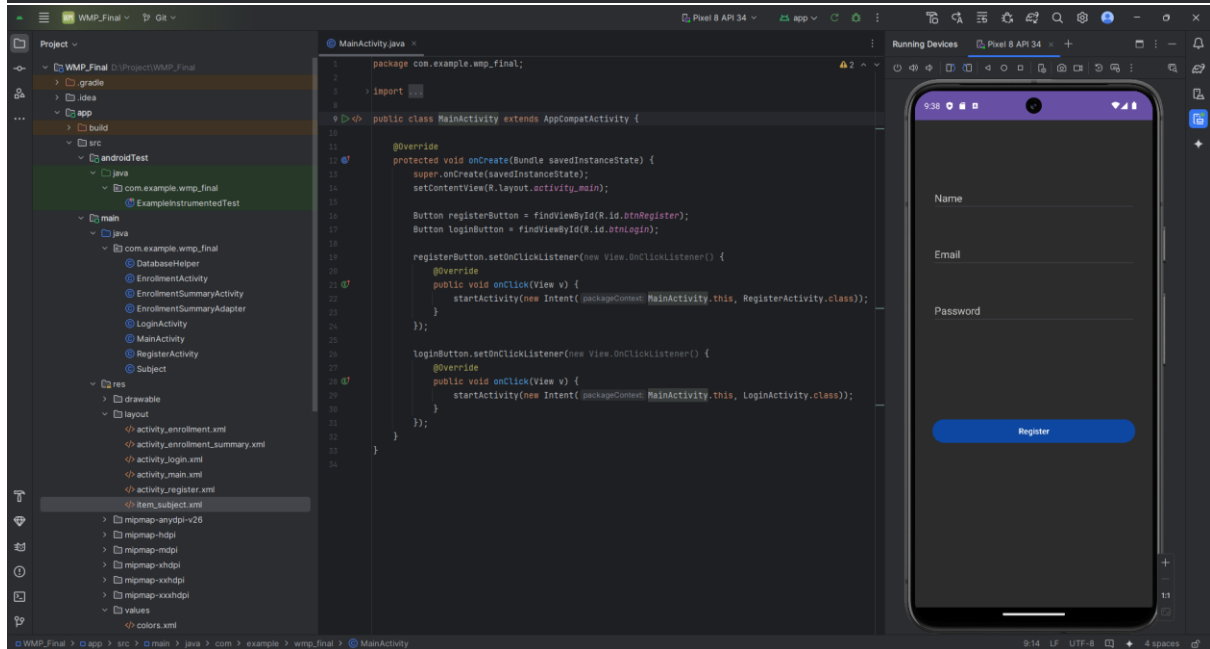
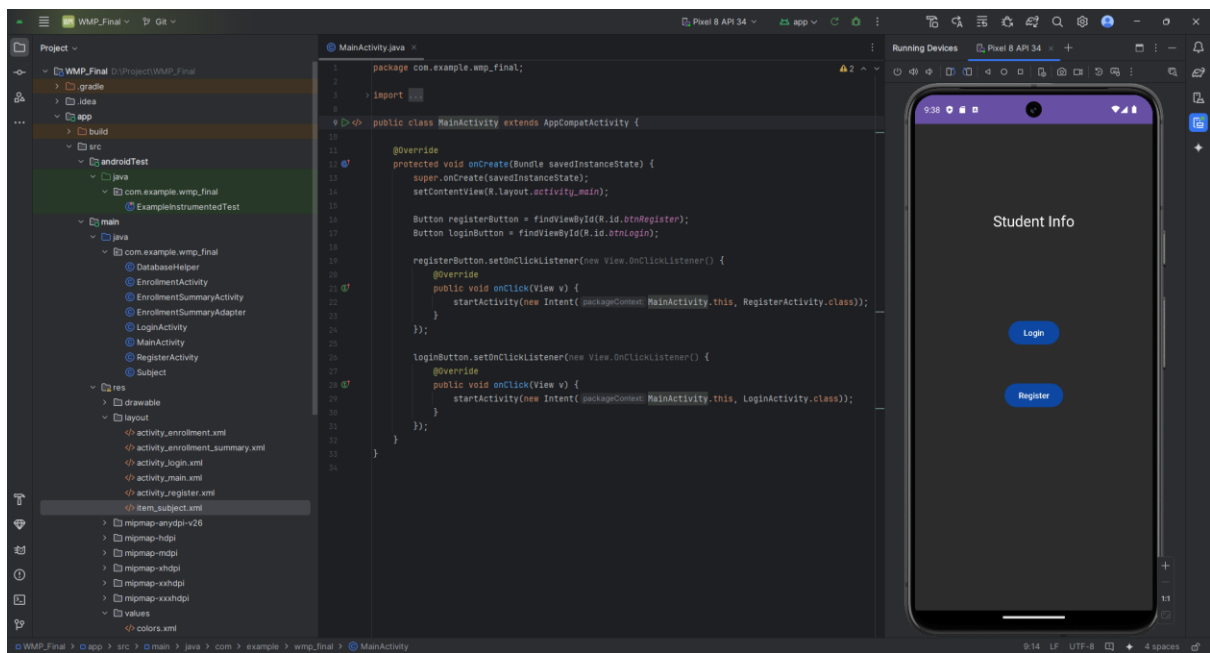
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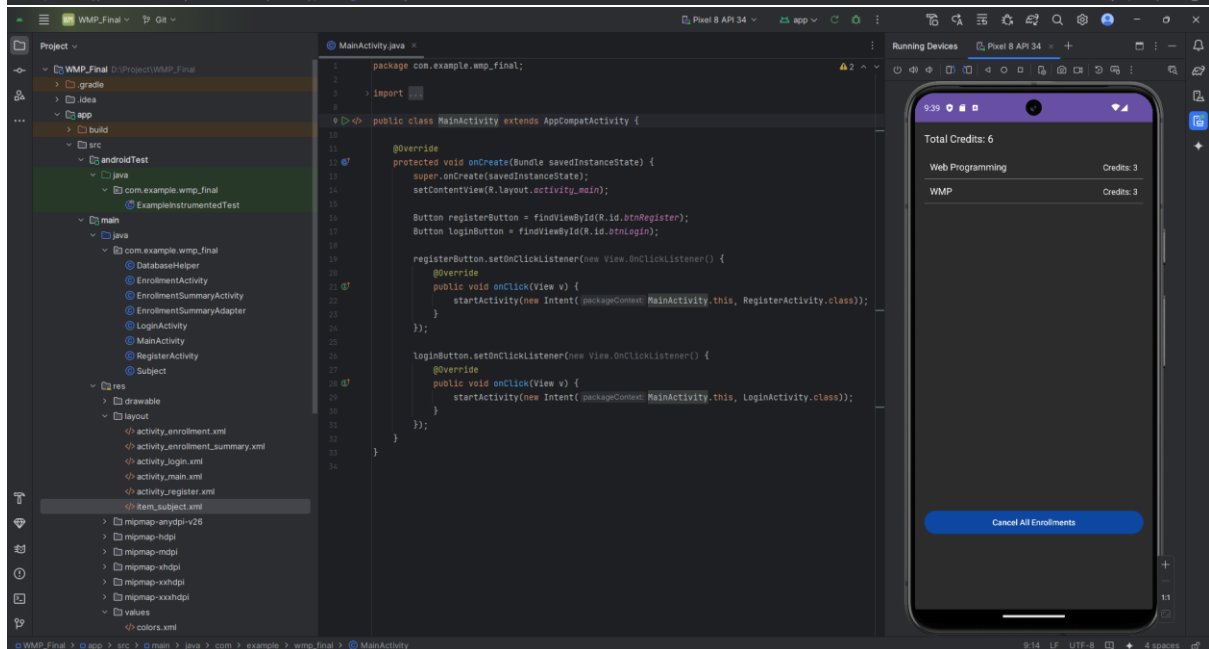
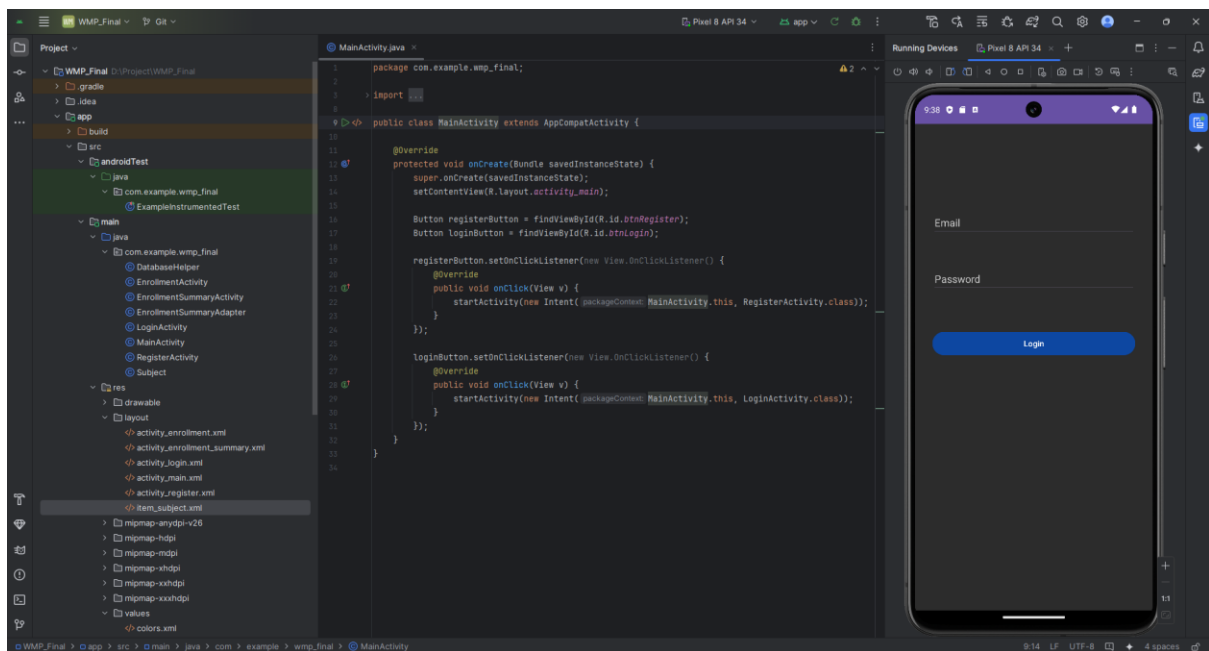
DatabaseHelper.java
1 package com.example.wmp_final;
2
3 import androidx.sqlite.db.SupportSQLiteOpenHelper;
4
5 public class DatabaseHelper extends SupportSQLiteOpenHelper {
6
7     8 usages
8     private static final String DATABASE_NAME = "school.db";
9     1 usage
10    private static final int DATABASE_VERSION = 1;
11
12    4 usages
13    public DatabaseHelper(Context context) {
14        super(context, DATABASE_NAME, null, DATABASE_VERSION);
15    }
16
17    @Override
18    public void onCreate(SQLiteDatabase db) {
19        db.execSQL("CREATE TABLE Student (" +
20            "student_id INTEGER PRIMARY KEY AUTOINCREMENT," +
21            "name TEXT NOT NULL," +
22            "email TEXT UNIQUE NOT NULL," +
23            "password TEXT NOT NULL);");
24
25        db.execSQL("CREATE TABLE Subject (" +
26            "subject_id INTEGER PRIMARY KEY AUTOINCREMENT," +
27            "subject_name TEXT NOT NULL," +
28            "credits INTEGER NOT NULL);");
29
30        db.execSQL("CREATE TABLE Enrollment (" +
31            "enrollment_id INTEGER PRIMARY KEY AUTOINCREMENT," +
32            "student_id INTEGER NOT NULL," +
33            "subject_id INTEGER NOT NULL," +
34            "FOREIGN KEY(student_id) REFERENCES Student(student_id)," +
35            "FOREIGN KEY(subject_id) REFERENCES Subject(subject_id));");
36
37        // Prepopulate subjects
38        db.execSQL("INSERT INTO Subject (subject_name, credits) VALUES ('Mathematics', 3), ('Robotics', 3), ('Automotive', 3), (" +
39            "'Informatics', 3), ('Sport', 3), ('Pharmacy', 3), ('Accounting', 3), ('Singing', 3), ('English', 3), (" +
40            "'Indonesian', 3), ('Web Programming', 3), ('WMP', 3);");
41    }
42}
```



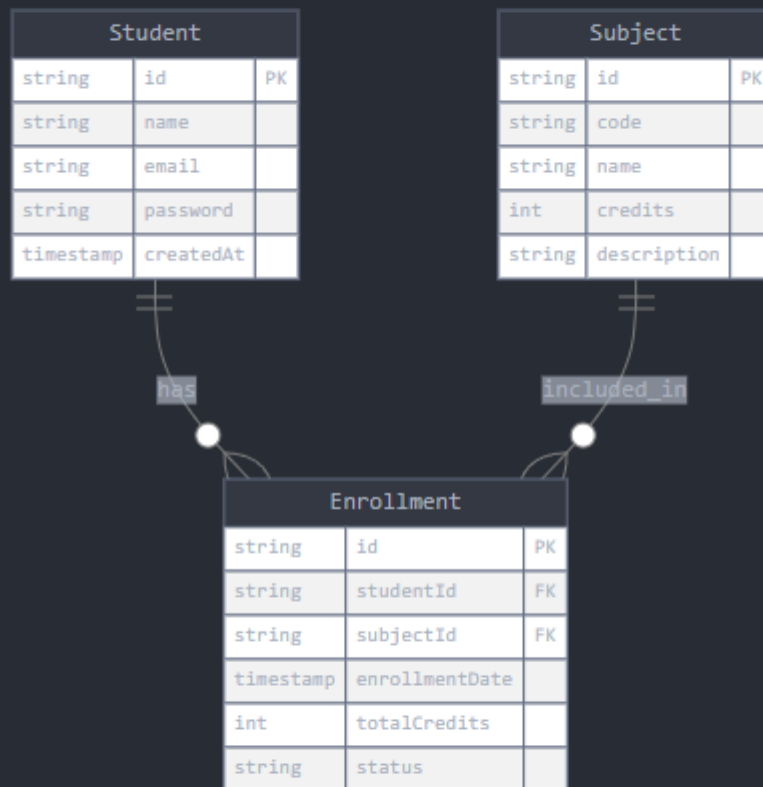








ERD



GitHub: https://github.com/PUTRAJAKBOYZ/WMP_Final-Exam